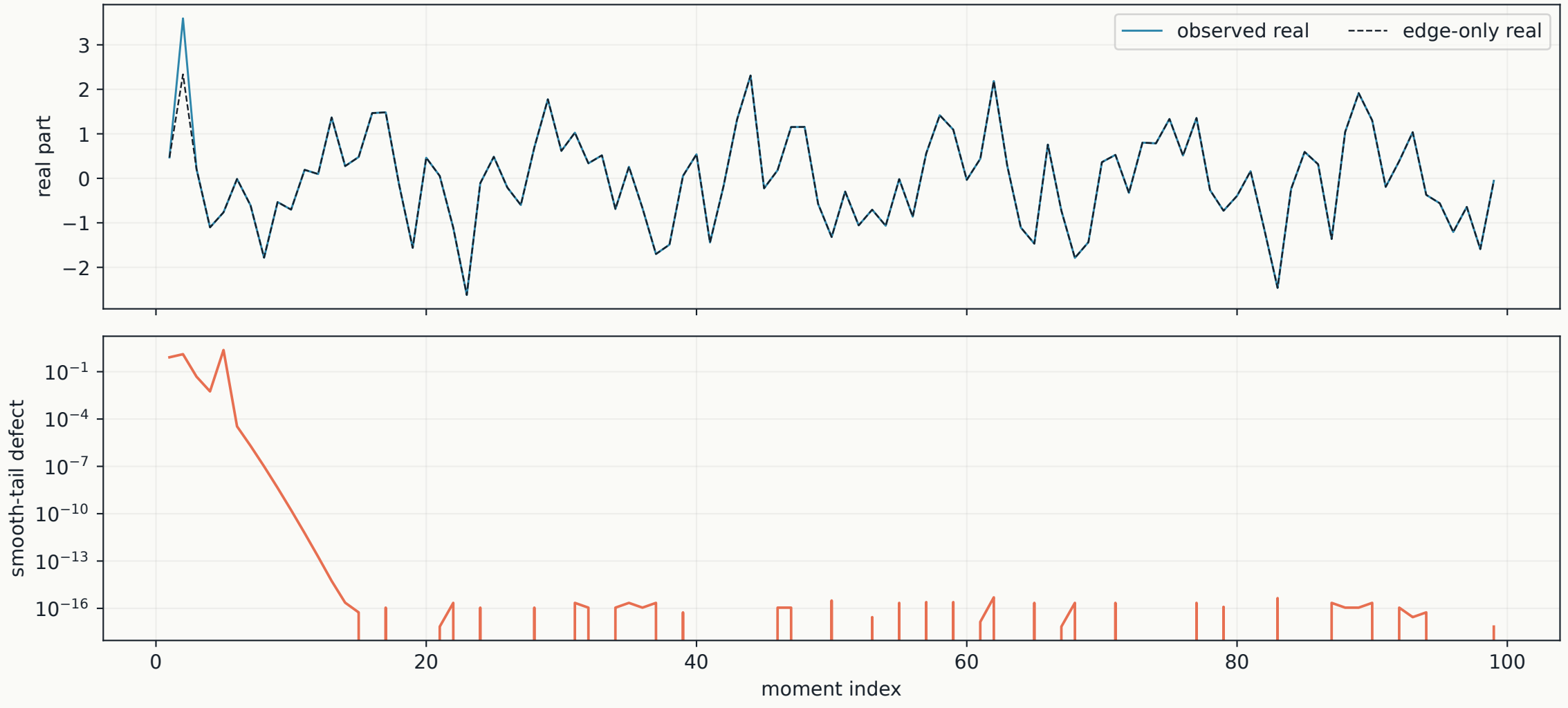Differentiation turns jumps into an exponential moment sequence